Problem 25.6

Given $\delta = 0.1$. Evaluate

$$\frac{a^{(12)}_{\overline{10}|}}{\overline{a}_{\overline{10}|}}.$$

Solution

We are given a constant force of interest

$$\delta = 0.1.$$

The annual effective interest rate i satisfies

$$\ln(1+i) = \delta.$$

Thus,

$$1 + i = e^{0.1},$$

so

$$i = e^{0.1} - 1 = 0.1051709181.$$

For payments made monthly, the nominal rate convertible monthly $i^{(12)}$ satisfies

$$1 + i = \left(1 + \frac{i^{(12)}}{12}\right)^{12}.$$

Therefore,

$$i^{(12)} = 12\left((1+i)^{1/12} - 1\right).$$

Since $1 + i = e^{0.1}$,

$$i^{(12)} = 12\left(e^{0.1/12} - 1\right) = 0.1004178265.$$

The present value of an annuity-immediate payable monthly for 10 years is

$$a^{(12)}_{\overline{10}|} = \frac{1 - v^{10}}{i^{(12)}}.$$

The present value of a continuous annuity for 10 years is

$$\overline{a}_{\overline{10}|} = \frac{1 - e^{-10\delta}}{\delta}.$$

Since $v^{10} = e^{-10\delta}$, the ratio is

$$\frac{a^{(12)}_{\overline{10}|}}{\overline{a}_{\overline{10}|}} = \frac{\frac{1-e^{-10\delta}}{i^{(12)}}}{\frac{1-e^{-10\delta}}{\delta}}.$$

Canceling $1 - e^{-10\delta}$,

$$\frac{a_{\overline{10}|}^{(12)}}{\overline{a}_{\overline{10}|}} = \frac{\delta}{i^{(12)}}.$$

Substitute $\delta = 0.1$ and $i^{(12)} = 0.1004178265$:

$$\frac{a_{\overline{10}|}^{(12)}}{\overline{a}_{\overline{10}|}} = \frac{0.1}{0.1004178265} = 0.9958391203.$$

Thus,

$$\boxed{0.9958391203}.$$